Seminar Report on

Insights to Business Intelligence & Analytics and Data Driven Culture

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Business Intelligence and Analytics (BI&A) has become an important part of the modern business world, driven by the increased availability of data and the need for real-time, evidence-based decision making. With the rapid growth of data sources and analytical technologies BI&A now supports a wide range of applications across industries.

1. Business Intelligence & Analytics

Business Intelligence (BI) includes the techniques, technologies, systems, practices, methodologies and applications that enable organizations to analyze business data so they can understand their business environment better and make decisions based on data (Chen, Chiang, & Storey, 2012). According to Delen and Ram (2018) BI also involves the discovery of meaningful patterns to find new insights that can guide actions in the organization. Analytics on the other hand means to support business objectives by reporting on trends, building predictive models, and optimizing business processes for improved performance (Delen & Ram, 2018).

A BI architecture typically integrates several components that were presented in the research paper from Chaudhuri, Dayal and Narasayya (2011): Data from diverse sources is prepared and moved using data movement and streaming engines, like Extract Transform Load (ETL) and Complex Event Processing (CEP) which support BI tasks in real time. The processed data is stored in data warehouses managed by relational database systems, while large-scale analytical workloads can be handled by MapReduce engines capable of executing complex SQL-like queries. Mid-tier servers like enterprise search engines, reporting servers or data and text mining engines provide specialized analytical functions. At the end of the architecture, insights are delivered to users through front-end applications such as dashboards, reporting tools, spreadsheets and ad hoc query systems, supporting flexible analysis and decision making. The architecture remained the same until today underlining the great functionality of this framework.

Chen, Chiang and Storey (2012) have introduced different technological fields of BI&A which are Big Data Analytics, Text Analytics, Web Analytics, Network Analysis and Web Analytics. Big data analytics combines traditional data warehousing with statistical machine learning based on mathematical models & powerful algorithms. Advanced algorithms, and large-scale distributed technologies such as Hadoop and MapReduce have been revolutionary for the field. Text analytics is a powerful extension of BI, enabling the extraction of structured information from unstructured documents (Chaudhuri, Dayal & Narasayya, 2011). It is mainly used for multimedia information retrieval and mobile information retrieval. Web analytics focuses on understanding user interaction patterns with web pages, facilitated by web search engines, cloud computing platforms, and scalable online infrastructures. It enables companies to understand how visitors interact with their web pages. Network analytics is grounded in bibliometric analysis and citation networks and relies on link mining and community detection. Finally, mobile analytics provides fine-grained, location-specific, and context-aware insights using data captured from smart devices and enables the usage of BI applications on mobile devices (Chen, Chiang and Storey, 2012). This shows the great amount of technologies that are being used for BI&A enabling the businesses to make better decisions based on actual data.

In their research paper Delen and Ram (2018) mention three different categories these technologies can be sorted by: descriptive, predictive, and prescriptive analytics. Descriptive analytics answers what has happened or is currently happening, often through reporting, dashboards, and data warehousing and is part of Business Intelligence. Predictive analytics asks what will happen and why, applying techniques such as data mining, text mining, web mining, and machine learning. Prescriptive analytics focuses on determining what should be done by employing optimization, simulation, decision modeling, and network science to support the best possible decisions. Both predictive and prescriptive analytics are part of advanced analytics.

1.1 Application Areas

BI&A is applied in many different industries all across the globe Delen and Ram (2018) mentioned some of them: In manufacturing, it supports order shipment and customer service; In retail, it drives pricing decisions, customer profiling, and customer relationship management. Financial services rely on BI&A for risk mitigation, fraud detection, and claims analysis. Transportation uses analytics for fleet management, while telecommunications companies apply it to understand customer churn.

Chen, Chiang and Storey (2012) go more into detail mentioning eight main application areas which are business and marketing, science and technology, healthcare and wellbeing as well as security and public safety. In business and marketing, BI&A supports website design, customer transaction analysis, market structure analysis, and product recommendation systems. All of these applications help organizations to monitor and to respond to the “voice of the market.” Social media analytics and sentiment analysis offer insight into how customers perceive brands. Health care benefits from BI&A through the analysis of electronic health records, decision support tools, and patient community analytics. Security and public safety applications include criminal association mining, crime analysis, terrorism informatics, and cyber-attack attribution. These different examples show that BI&A is used in almost every industry shaping our life. A lot of users are confronted with BI&A everyday without even noticing it.

1.2 Evolution

The evolution of BI is divided in three different stages by Chen, Chiang and Storey (2012). BI&A 1.0 is rooted in structured data management and relies heavily on ETL tools, database queries, online analytical processing (OLAP) and simple reporting tools. This period was there from the 1970s until the late 1980s. Scorecards and dashboards promote business performance management and help visualize key metrics (Delen & Ram, 2018).

The second stage BI&A 2.0 started with the rise of web-based, unstructured content in the early 1990s. The large-scale data collection and direct user interaction through the web make it possible to capture feedback and opinions from different customers. Technologies such as opinion mining and information retrieval became more and more important and data warehousing became a standard across medium and large organizations. The decision making shifted from intuition to evidence-based decision making (Delen & Ram, 2018).

BI&A 3.0, the last stage in the research paper from Chen, Chiang and Storey (2012) started in the 2010s when mobile technologies, IoT, and sensors became more popular. These tools

increased the amount of high-quality organizational data. New parallel data architectures and analytic tools have enabled almost real-time BI (Chen, Chiang, & Storey, 2012). The competitive pressure has led to the increased need for near real time BI further pushing the demand to deliver interactive BI experiences on mobile devices for knowledge workers (Delen & Ram, 2018)

Today, AI-powered BI tools are embedded across organizational functions, enabling intelligent operations, real-time decision making, high-accuracy predictions, and increasingly autonomous systems. In the future natural language processing, computer vision, and reinforcement learning will enable BI systems to understand context and keep continuously improving (Henkin 2024).

This evolution of BI&A underlines clearly how the way we do business is strongly shaped by technological advancements and that businesses are able to adapt and use new technologies to their advantage. How can such a culture be integrated in an organization and what are the challenges and opportunities?

2. Data Driven Culture

Creating a data-driven culture means fostering an organizational environment where data is consistently collected, analyzed, and used to guide decisions. It is shaped by a mindset that views data as a critical asset rather than an optional resource to guide decisions on all levels of the organization (Davenport & Malone 2021; Berndtsson & Ekman 2023). Berndtsson and Ekman (2023) write in their research paper that teams are encouraged to ask analytics-driven questions and that leadership commitment is essential for having a data-driven culture.

Data-driven culture is not a one-time deployment task but a continuous learning and development cycle (Davenport & Malone, 2021). Successful adoption depends on change management and trust in automated decisions and not just on the accuracy of analytical models but only about one-third of organizations successfully transition to a data-driven culture (Berndtsson & Ekman, 2023). What are the challenges that hold some companies back?

2.1 Problems

Organizations face significant challenges in achieving a Data Driven Culture. Legacy systems often have a negative impact on modern data practices, as outdated architectures struggle with unstructured data and lack real-time processing capabilities (Davenport & Malone, 2021). Deployment is another major challenge: despite investing a lot of money in data science, very few organizations deploy models they are developing. Only a small percentage of models reach production, and even fewer bring significant financial benefits. Data inconsistencies with training data or unreliable and changing data are often the problem at deployment (Davenport & Malone, 2021).

Non-technical barriers include insufficient management support, skill shortages, employee resistance, poor data quality, and a lack of data literacy. Organizational culture is often not aligned with analytics, and changing established behaviours is a difficult challenge (Berndtsson & Ekman 2023; Kaushik 2007). Berndtsson and Ekman (2023) also highlight that exist-

ing maturity models for analytics don't address group development. This should be implemented in models since data-driven culture completely changes group dynamics since it is shifting from an intuition-based decision system to an evidence-based one.

2.2 Opportunities

Although there are challenges to overcome there are also great opportunities that lie in a data-driven culture for example do organizations gain significant competitive advantages. Industry leaders such as Google and Amazon are a great example for the potential of data-centric business models (Berndtsson & Ekman, 2023). Effective data practices improve measurement, forecasting, decision quality, and operational productivity (Davenport & Malone 2021). Analytics also strengthens team dynamics, enabling discussions based on evidence rather than anecdotes. Investments in group development help prevent “territorial analytics” which means that analytics is used for some applications in the company but the findings are not actively communicated with the other people in the company. (Berndtsson & Ekman 2023). This shows that there is great potential in a data-driven culture as long as it is integrated in every part of the organization.

2.3 Successful Implementation

Davenport and Malone (2021) outlined a systematic approach to successful BI&A that begins with a clear understanding of the business problem at hand in the real world process and to be aware of the potential failure modes of analytical models. It is also crucial to building trust with key stakeholders. Last but not least organizations must understand the technical environment in which models will operate, establish accountability for results, and continuously monitor performance after deployment. Effective implementation depends on multidisciplinary teams, hiring talent with deployment experience rather than modeling skills alone, and ensuring that every analytics project includes a robust deployment pipeline. Fostering data literacy is crucial as well as creating incentives for data-driven decision making, depersonalizing analytical outcomes, and developing strong customer experience analytics to guide decisions (Davenport & Malone, 2021; Kaushik, 2007).

3. Conclusion

BI&A has turned into a powerful set of tools and practices that enables organizations to make informed, data-driven decisions. Technological advancements have expanded analytical capabilities, but the true value is only achieved when organizations also develop a data-driven culture. Organizations that overcome cultural and organizational barriers gain clearer insights, improved decision quality, and a stronger competitive position.

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